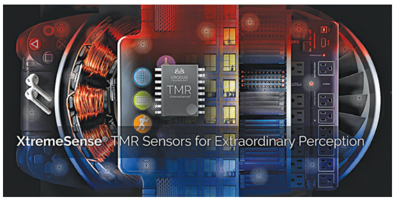
offers high integration, and best-in-class power consumption. It has the flexibility to address multiple optical transceivers and end markets. Keystone DSPs can address both 400G and 800G optical module applications.

MaxLinear

www.maxlinear.com

Tunnel magneto-resistance technology for various end products

Crocus Technology has developed an innovative XtremeSense tunnel magneto-resistance technology to provide superior magnetic sensing performance. It enables high magnetic sensitivity, stable performance over temperature, low noise, and low power consumption, which allows it to be used in a wide



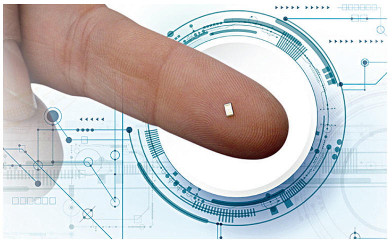
variety of end-products and markets. At the heart of this technology is a thin-film magneto-resistive device, the magnetic tunnel junction, which consists of two electrically-conducting magnetic layers on either side of a thin nanometre-scale but highly robust insulating layer. One magnetic layer has a fixed magnetic moment direction while the other can change freely to follow the direction of the local magnetic field.

Crocus Technology

www.crocus-technology.com

World's smallest 3D ultrasound sensor for standalone operation

UltraSense has developed the world's smallest and most integrated 3D ultrasound sensor, ushering a new era for user interfaces to be reimagined. Now, virtually any material with any thickness can become a button or gesture control. Target applications include smartphones, mobile devices, home appliances, automotive, medical, and industrial. Designed for standalone operation, the sensor includes a transducer



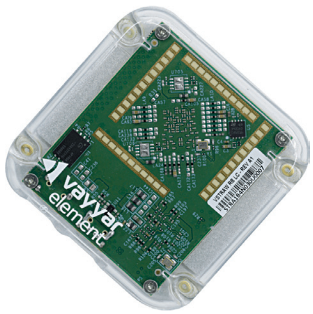
and an ASIC in a monolithic silicon die. Its features include a multifunctional interface, and ability to distinguish touch input materials such as finger, latex, cloth, etc. It is waterproof.

UltraSense Systems

www.ultrasensesys.com

4D imaging radar with unprecedented resolution

Developed to detect and track people in real time, this miniature 4D imaging radar called Vayyar Element has the ability to communicate with everyday products using 4D imaging radar technology. The module uses leading-edge



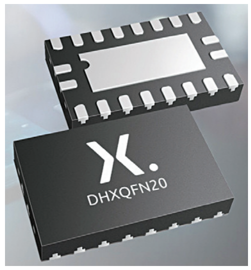
imaging radar technology to generate real-time point cloud images and provide location data of people in its vicinity. With a large array of 46 RF transceivers, Vayyar Element delivers unprecedented radar resolution. Unlike optics based solutions, imaging radar ensures privacy by design and does not collect personal data, supporting GDPR compliance. It can be used in all lighting and environmental conditions including darkness, smoke, steam, glare, and fog.

Vayyar

www.vayyar.com

Small and thin logic IC for edge devices

Nexperia, a provider of semiconductor solutions, has announced small and low-profile 14, 16, 20, and 24-pin packages for standard logic devices. The 16-pin DHXQFN package is 45% smaller than the industry-standard DQFN16 leadless device. Not only does it have a smaller footprint, it also offers 25% saving in



PCB area. Measuring just 2mm x 2mm (14-pin), 2mm x 2.4mm (16-pin), 2mm x 3.2mm (20-pin), and 2mm x 4mm (24-pin), the 0.4mm pitch packages are only 0.45mm high. Parts include hex inverting Schmitt-triggers, 8-bit SIPO shift registers with output latches, 4-bit dual supply translating transceivers, octal buffer/line drivers, octal bus transceivers, and 8-bit dual supply translating transceivers.

Nexperia

www.nexperia.com

Low-current microphone for always-on applications

TDK Corporation introduces the InvenSense T5838, the latest technology breakthrough in the SmartSound product family. It is a low-power pulse density modulation, multi-mode MEMS

microphone with high AOP and high SNR for smartphones, microphone arrays, smart speakers, IoT, and other consumer devices. The microphone offers an exceptionally efficient 130µA ultra-low power mode to support always-on applications. The T5838 has outstanding low current of 330µA in high-quality mode, supporting high fidelity audio with 68dB SNR and 133dB AOP at 20kHz bandwidth in a 3.5mm x 2.65mm x 0.98mm package.

TDK Corporation

www.tdk.com